



FACULTY OF COMPUTER SCIENCE AND MATHEMATICS
BACHELOR OF COMPUTER SCIENCE
(SOFTWARE ENGINEERING) WITH HONOURS

CSE3433 SOFTWARE ARCHITECTURE

SEMESTER II SESSION 25/26

PROJECT TITLE:

RECRUITX - EVENT-DRIVEN JOB RECRUITMENT & APPLICATION
PROCESSING SYSTEM

PREPARED FOR:

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Selected Architecture Style

The RecruitX system will implement the Event-Driven Architecture using the concept of Publish/Subscriber model. Instead of rely on the traditional request, the system operates through the generation and consumption of the “Events”.

The component is:

- Event Consumers: Independent service such as Email Notifications and Real Time Analytics that “subscribe” to the bus and execute tasks automatically once the event is detect.
- Event Producer: The user interfaces for Candidate and Recruiters. They “publish” an event notification to the system when an action is performed.
- Event Bus: The central hub that receive events and routes them to the appropriate services. It acts as a buffer, ensuring the system remain stable during high-traffic hiring session.

Our group choose this because:

- Decoupling: Service functions independently. If the service is down, the application submission process still can use.
- Scalability: The Event Bus manages bursts of incoming resumes efficiently, preventing the system crashes during the peak periods.

Problem Description

The current recruitment is heavily manual and slow, which causes several problems for the organization:

- Slow hiring process: HR teams must read through every resume and contact candidates one by one. This takes a lot of time and effort, making the whole process move very slowly
- No immediate action: When a candidate submits an application, nothing happens automatically. The application just sits there until an HR staff member manually opens it, leading to long delays in shortlisting
- Poor communication: Candidates often wait a long time for updates. Since communication is slow, top talent may lose interest or accept jobs from other companies before they hear back
- Risk of mistakes: Handling everything by hand makes it easy for humans to make errors.
- Difficulty handling many applicants: If a job gets a large number of applications, the manual system becomes overwhelmed. The team cannot work fast enough to keep up with the workload.

System Objectives

The objectives of RecruitX are:

1. To automate the entire process of recruitment by initiating actions from the system upon occurrence of certain events, such as job posts, applications submissions, and scheduling interviews.
2. To decrease the effort needed to be put forth by HR personnel in recruiting employees by automating the resume filtering process, managing applications, and executing other repetitive tasks of recruitment.
3. To ensure efficient communication among candidates and recruiters with the use of real-time notifications and automatic updates regarding applications.
4. To eliminate any sort of human errors during the entire recruiting process by managing all activities involved in it in an organized way.
5. To make quick decisions for hiring new employees using the automation of short-listing and scheduling of interviews according to certain criteria.
6. To generate real-time recruitment insights and reports that help organizations monitor hiring performance and improve decision-making.

Initial Functional Requirements

FR01 - The system shall allow all the recruiters to create and publish the job listings which is shall trigger a JobPosted event to update the public job board automatically.

FR02 - The system shall, when submission of an application, trigger an ApplicationReceived event to provide the candidate with an intermediate automated confirmation.

FR03 - The system shall trigger an ApplicationStatusChanged event when a recruiter update all the candidate progress, automatically sending an update to the candidate to eliminate the communication delays.

FR04 - The system shall automatically screen the incoming resumes based on pre-defined keywords like skills or year of experience to assist HR in narrowing down to the applicant pool without manual effort.

FR05 - The system shall provide a centralized interface for recruiters to move the candidate through stages like shortlisted or interviewed, ensuring no qualified candidate is overlooked.

FR06 - The system shall allow the recruiters to initiate the InterviewRequested event which is automatically coordinated with the candidate's availability and sends the calendar invites.

FR07 - The system shall provide a dashboard that monitors the "hiring funnel" in real-time, reducing all the human error in data entry or tracking.

Team Member Roles

Name	Role	Responsibilities
Nur Alin Hayani binti Ahmad Syukri	System Architect & Designer	Responsible for UML Architecture Diagrams, Justify the Architectural Drivers
Ainaa Nadhirah binti Asmadi	Backend & Integration Lead	Backend service development, RabbitMQ configuration, database management, and API implementation.
Maryam Musfirah binti Ariff Hazizan	Requirements Analyst	Analyzing the problem statement, defining system objectives, preparing FR, and identifying architectural drivers.